

JINGMING YAN

jingmy1@uci.edu

<https://jingming-yan.github.io>

EDUCATION

University of California, Irvine

September 2023 - Present

Ph.D. in the department of Computer Science

- Advised by Prof. Ioannis Panageas
- My research lies at the intersection of Algorithmic Game Theory, Min-Max Optimization, Reinforcement Learning, and Computational Complexity

University of California, Irvine

September 2019 - June 2023

B.S. in the department of Mathematics

- GPA: 4.0/4.0
- Minor: Information and Computer Science

RESEARCH INTERESTS

Min-Max Optimization, Non-Convex Optimization, Complexity Theory, Stochastic Games, Multi-Agent Reinforcement Learning, Learning in Games

PUBLICATIONS

1. **The Complexity of Symmetric Equilibria in Min-Max Optimization and Team Zero-Sum Games**
Ioannis Anagnostides, Ioannis Panageas, Tuomas Sandholm, **Jingming Yan**
NeurIPS 2025 Spotlight
2. **The Complexity of Finding Local Optima in Contrastive Learning**
Jingming Yan, Yiyuan Luo, Vaggos Chatziafratis, Ioannis Panageas, Parnian Shakhar, Stelios Andrew Stavroulakis
NeurIPS 2025
3. **Learning Equilibria in Adversarial Team Markov Games: A Nonconvex-Hidden-Concave Min-Max Optimization Problem**
Fivos Kalogiannis, **Jingming Yan**, Ioannis Panageas
NeurIPS 2024
4. **On the Computational Complexity of Performative Prediction**
Ioannis Anagnostides, Rohan Chauhan, Ioannis Panageas, Tuomas Sandholm, **Jingming Yan**
Arxiv preprints

EXPERIENCE

Research Intern at Archimedes Research Unit

June 2025 - August 2025

Athens, Greece

- Conducted research on designing efficient methods and analyzing the computational complexity of solving variational inequalities in high-dimensional domains
- Developed beyond-worst-case dynamics for local updates to ensure convergence in contrastive learning settings, and analyzed their convergence rates
- Publications got accepted in NeurIPS 2025

Research Intern at Archimedes Research Unit
Athens, Greece

June 2024 - August 2024

- Researched on intractability and complexity of min-max optimization and team games
- Generated stimulated converging dynamic for first-order algorithms and applied in various settings

Undergraduate Research Experience

September 2022 - June 2023

Supervisor: Prof. Ioannis Panageas

- Analyzed the effect of applying different regularizers in min-max optimization
- Systematically learned algorithmic game theory, stochastic games, min-max optimization tools (e.g. Optimistic GDA), nonconvex optimization techniques (e.g. Moreau Envelope)
- Implemented code that studied the convergence in stochastic two-player zero-sum games and stochastic potential games

ACADEMIC SERVICES

Reviewer for: AISTATS 2024, 2025, 2026; AAAI 2025; ICML 2024, 2026; NeurIPS 2024, 2025; ICLR 2025, 2026

TEACHING EXPERIENCE

COMPSCI-161 Design & Analysis of Algorithms	<i>Winter 2026</i>
COMPSCI-161 Design & Analysis of Algorithms	<i>Spring 2025</i>
COMPSCI-178: Machine Learning and Data Mining	<i>Winter 2025</i>
ICS-46: Data Structure Implementation and Analysis	<i>Fall 2024</i>
COMPSCI-161 Design & Analysis of Algorithms	<i>Winter 2024</i>
ICS-46: Data Structure Implementation and Analysis	<i>Fall 2023</i>
COMPSCI-161 Design & Analysis of Algorithms	<i>Spring 2023</i>

SKILLS

Programming : C++, Python, Pytorch, TensorFlow, MATLAB, Mathematica, R, $\text{\LaTeX}$

Languages : Chinese (native), English